

Lecture 8

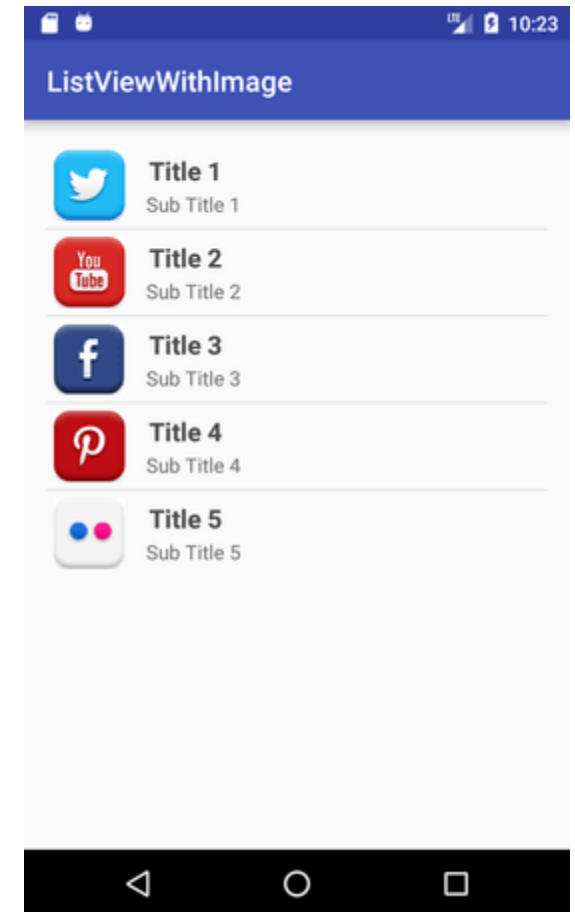
List Views and Base Adapter

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Base Adapter

- BaseAdapter is a common base class of a general implementation of an Adapter that can be used in ListView, GridView, Spinner etc.
- Whenever you need a customized list in a ListView or customized grids in a GridView you create your own adapter and extend base adapter in that.
- Base Adapter can be extended to create a custom Adapter for displaying a custom list item.



CustomBaseAdapter.java

```
package com.example.listviewexample;

import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;

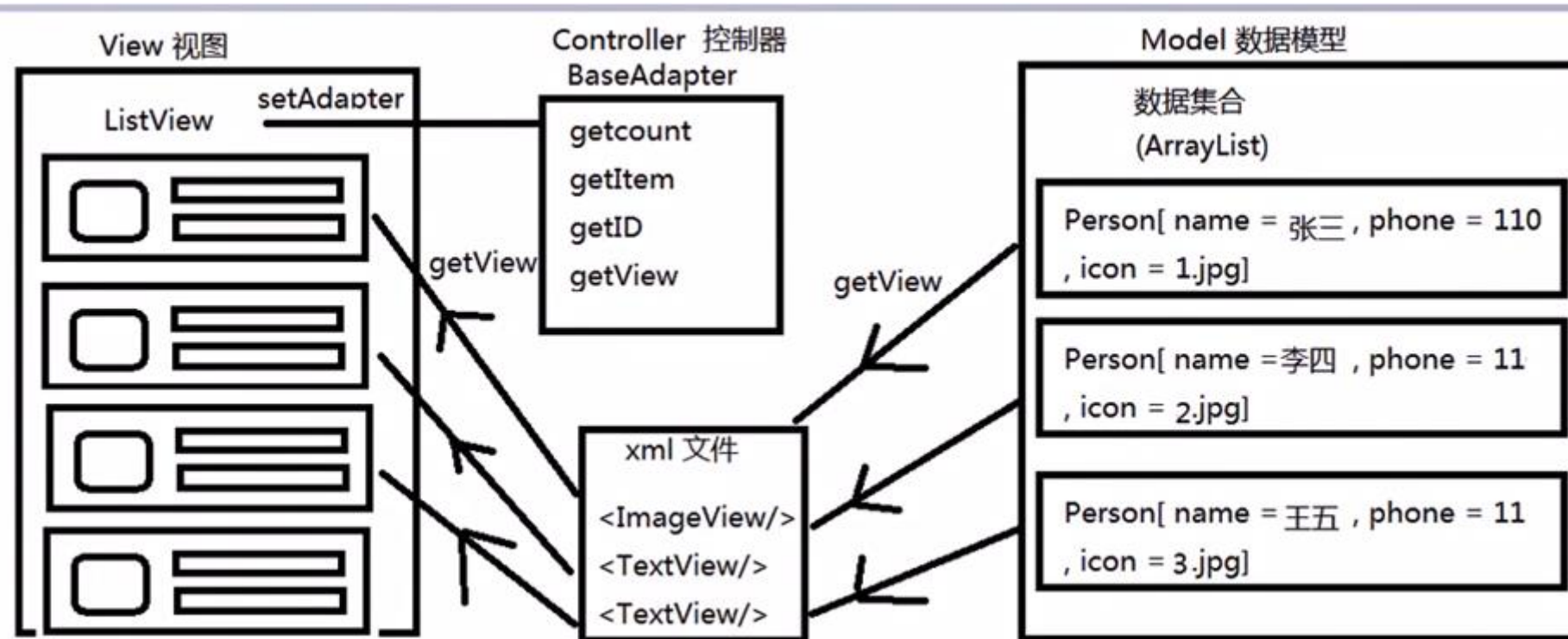
public class CustomBaseAdapter extends BaseAdapter {
    @Override
    public int getCount() {
        return 0;
    }

    @Override
    public Object getItem(int position) {
        return null;
    }

    @Override
    public long getItemId(int position) {
        return 0;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        return null;
    }
}
```

Base Adapter Working



Base Adapter Function

- getCount()
 - The getCount() function returns the total number of items to be displayed in a list.
- getView(int position, View convertView, ViewGroup parent)
 - This function is automatically called when the list item view is ready to be displayed or about to be displayed.
 - Get a View that displays the data at the specified “position” in the data set.
 - In this function we set the layout for list items using LayoutInflater class and then add the data to the views like ImageView, TextView etc.
 - LayoutInflater Instantiates a layout XML file into its corresponding view object

Base Adapter Function Cont.

- getItem(int i)
 - To get item position of the item. E.g. `arrayList.get(i);`
- getItemId(int i)
 - To get id of the item which is most of the time is index of the item in the list

Custom List View's Item Layout

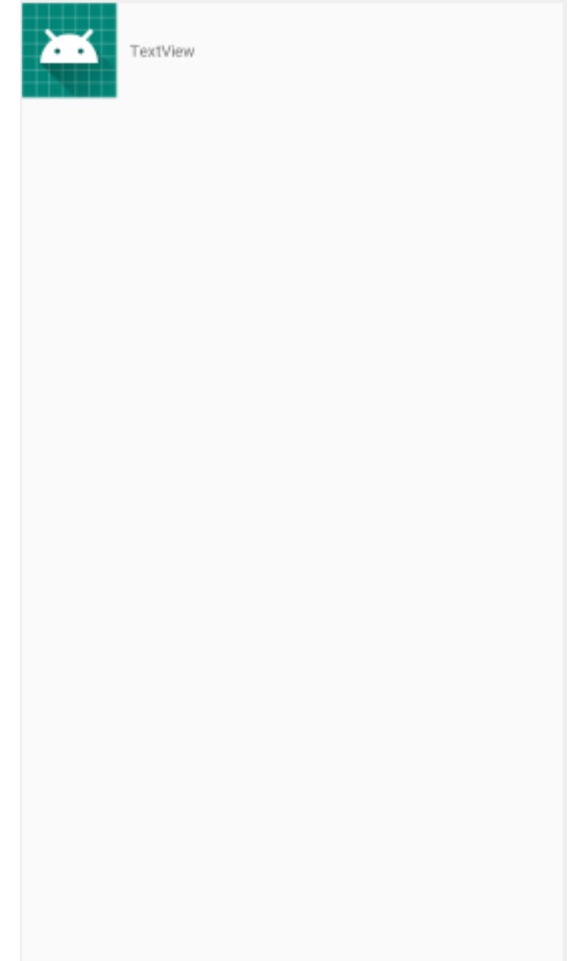
custom_list_view_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="horizontal" android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <ImageView android:id="@+id/list_item_image"
        android:src="@mipmap/ic_launcher"
        android:layout_height="wrap_content"
        android:layout_width="wrap_content">
    </ImageView>

    <TextView
        android:id="@+id/list_item_text"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="TextView"
        android:layout_marginLeft="10dp"
        android:gravity="center_vertical"
        android:textSize="12dp" />

</LinearLayout>
```



```

public class CustomBaseAdapter extends BaseAdapter {

    Context context;
    String[] OSName;
    int[] OSImageLoc;
    LayoutInflater inflater;

    public CustomBaseAdapter(Context context, String[] OSName, int[] OSImageLoc){
        this.context = context;
        this.OSName = OSName;
        this.OSImageLoc = OSImageLoc;
        inflater = LayoutInflater.from(context);
    }
    @Override
    public int getCount() {
        return OSName.length;
    }

    @Override
    public Object getItem(int position) {
        return OSName[position];
    }

    @Override
    public long getItemId(int position) {
        return position;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        // get view of custom item layout
        convertView = inflater.inflate(R.layout.custom_list_view_item, null);

        ImageView OSImageView = (ImageView) convertView.findViewById(R.id.list_item_image);
        TextView OSText = (TextView) convertView.findViewById(R.id.list_item_text);

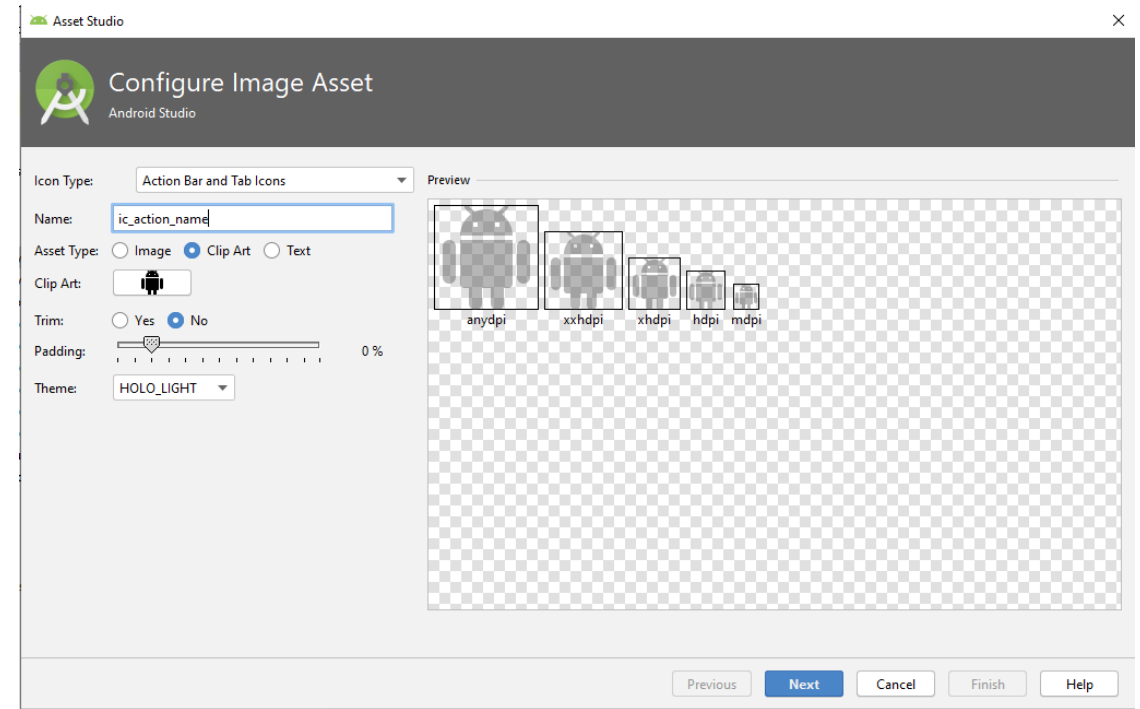
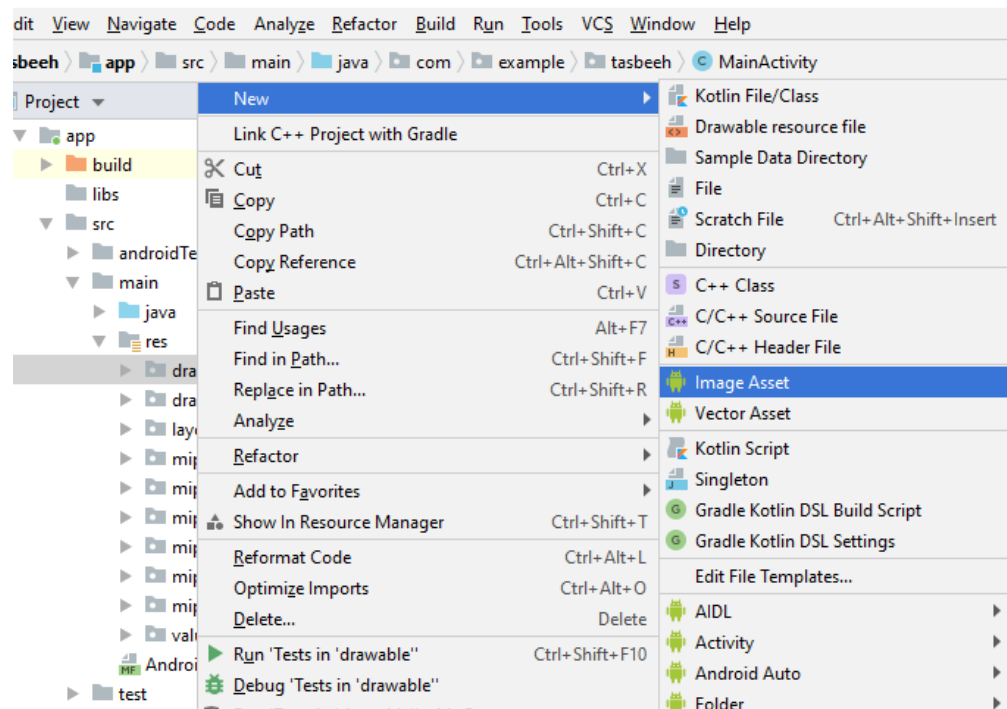
        OSImageView.setImageResource(OSImageLoc[position]);
        OSText.setText(OSName[position]);
        return convertView;
    }
}

```

Custom Adapter Class
CustomBaseAdapter.java

Add Images Assets in Drawable

- To add new images in drawable folder, use image assets to add new image from repository or from local storage



MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    String[] OSName = {"Android", "iOS", "Windows"};
    int[] OSImageLoc = {R.drawable.android_icon, R.drawable.ios_icon, R.drawable.window_icon};

    ListView operatingSystemList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        operatingSystemList = (ListView) findViewById(R.id.mobile_list);
        operatingSystemList.setAdapter(new CustomBaseAdapter(getApplicationContext(), OSName, OSImageLoc));
    }
}
```

